Structural Damage Image Classification — A Reproduction Study

UE24CS352A Machine Learning — Mini-Project | Sanjith K S | Reproducing M. Ho and J. Troncoso, *Structural Damage Image Classification*, CS229 Final Project, Stanford University, 2018.

1. Problem statement

Given a photograph of a structure, decide whether it shows damage. The motivation is rapid post-earthquake assessment: after a seismic event, engineers must triage very large numbers of buildings quickly, and an automated classifier applied to field photographs could prioritise inspection. The task is binary image classification on a small, visually heterogeneous dataset — images range from close-ups of a single hairline crack to wide shots of entire apartment blocks, and contain irrelevant objects such as people, vehicles, curtains and tree branches. Our objective was to reproduce the six models reported by Ho and Troncoso, confirm or refute their conclusions, and account for any differences.

2. Dataset

PEER Hub ImageNet (Φ-Net), Task 2 — Damage State, released under CC BY-NC-SA 4.0. The paper used 5,913 labelled 224×224 RGB images (3,186 undamaged, 54%; 2,727 damaged, 46%) split 90/10 into 5,321 training and 592 validation images. The public release is now larger (13,271 images), so we subsampled it to the paper's exact size and per-class counts under a fixed seed, giving splits identical in composition to theirs. The paper deliberately kept no test set; we additionally evaluate on the official 1460-image Φ-Net test split, which it never used.

Two undocumented properties of the distribution had to be recovered empirically. First, the images are **not raw pixels**: they are stored in BGR channel order with the ImageNet per-channel mean already subtracted, giving a range of −123.68 to +151.06. Adding the means back and reversing the channel axis recovers exact 8-bit RGB; clipping to [0,255] without this step destroys roughly half of every image while raising no error. Second, the shipped label convention (0 = damaged) is the **inverse** of the paper's (0 = undamaged), which we inferred from the row sums of its confusion matrix. We adopted the paper's convention and verified the choice by direct visual inspection, since a label flip leaves every accuracy metric unchanged and corrupts only the interpretation. Inspection also confirmed the paper's remark that some images are mislabelled: one in a random sample of sixteen showed a spalled bridge girder with exposed reinforcement labelled *undamaged*, which places a ceiling on achievable accuracy.

3. Approach

We reimplemented the paper's models in PyTorch; the authors' 2018 TensorFlow 1.x code no longer runs and was used only as a specification. Three classical models operate on raw pixels flattened to 150,528-dimensional vectors scaled to [0,1]: k-nearest neighbours (k=5), logistic regression (L2, C=1.0) and an RBF support vector machine (C=1.0, γ=0.001). The deep model applies transfer learning: InceptionV3 pretrained on ImageNet with every layer frozen, its 1000-way classifier removed, and a single fully-connected layer with softmax trained on the 2048-dimensional pooled features using plain gradient descent (lr 0.01, batch 100, 4,000 iterations) — the authors' hyperparameters throughout. We then reproduced their two diagnostics: a learning curve over training-set size, and a confusion matrix with galleries of the most confident misclassifications.

4. Results

Model	Paper train	Paper val	Our train	Our val
K-nearest neighbours (k=5)	67.4%	53.7%	71.3%	50.8%
Logistic regression	99.4%	54.7%	99.8%	54.0%
SVM (RBF), paper's 3,000-image protocol	99.7%	50.7%	95.1%	55.9%
InceptionV3, frozen + FC/softmax	81.0%	83.0%	88.5%	79.6%
— same, 299×299 input (ours)	—	—	87.6%	81.8%
— SVM on all 5,321 images (ours)	—	—	94.9%	57.8%
PRODUCTION: all 11,219 images, 299px	—	—	—	82.3%

Majority-class baseline on the validation set: 53.4%. Shaded row: the paper's headline model. Cream rows: experiments beyond the paper. Held-out test accuracy, for which the paper has no counterpart: **81.2%** for the reproduction and **82.7% ± 0.5** for the production model, on 1460 images.

The paper's central claim reproduces cleanly: classical models on raw pixels are at best a couple of points above a constant guess despite 95–99% training accuracy, while transfer learning reaches roughly 80%. With 150,528 features and 5,321 examples the models have 28 times more parameters than data points, so memorisation is free and generalisation is not; an RBF kernel of width γ=0.001 over these distances is in addition far too narrow to generalise. One methodological note: the saved splits are ordered by class, so slicing the first N rows is not a sample. All subsampling here is stratified under a fixed seed, and the scripts report the class balance actually trained on.

Our InceptionV3 validation accuracy (79.6%) falls below the paper's reported 83.0%, but matches the **79.7%** implied by the paper's own confusion matrix in its error-analysis section to within 0.1 points. The paper's Table 1 figure also has validation accuracy exceeding training accuracy, which a frozen backbone with a 4,098-parameter head should not produce; ours is the conventional way round. We additionally found that the paper fed 224×224 images to a network pretrained at 299×299.

Our learning curve **contradicts the paper's diagnosis**. The paper read its curve as "clearly a bias issue" and responded by unfreezing layers to add capacity. Averaged over three seeds, we measure a train/validation gap of 26 points at 200 training images, falling to 8.6 points at 5321 — the signature of variance, not bias. At full size the validation curve has flattened near 80%, so further data will not help; the binding constraints are the frozen ImageNet features and label noise rather than model capacity. ImageNet contains only 1,190 structural images out of 14.2 million, so its top-layer features are a poor match for this domain.

Error analysis shows the failures are structural rather than random. False positives are dominated by high-texture undamaged surfaces — brick courses, tiling, tree branches — and by genuinely mislabelled images where the model is in fact correct. False negatives are dominated by wide shots in which damage is signalled by context, such as shoring props, scaffolding, boarded windows or rubble, rather than by crack texture. The model has learned close-up crack appearance. This confirms the paper's hypothesis that "patterns can be mistaken as cracks, and vice versa", and demonstrates the close-up versus wide-angle distinction that the paper listed only as future work.

Having reproduced the paper at its own settings, we then removed its two incidental limitations. The dataset has more than doubled since 2018, and InceptionV3 expects 299×299 inputs rather than the 224×224 the paper fed it. Training the identical head on all 11,219 available images at 299×299 gives **82.7% ± 0.5** on the held-out test set over 3 seeds, with damaged-class recall of 81.7% against the paper's 74.9%. For a triage application that recall is the figure of merit: it is the fraction of damaged structures the system does not miss. We note that validation accuracy varied by 1.7 points across seeds while test accuracy varied by 1.1, so any single-run figure on a set this size carries at least a point of noise - which may explain the paper's 83.0%.

5. Implementation

All code is PyTorch and scikit-learn, run on a single 6 GB laptop GPU. Two engineering decisions made the study feasible. First, because the backbone is frozen its output per image never changes, so the 2048-dimensional features are extracted once and cached; the 4,000 training iterations then become matrix multiplies on a 5321×2048 array, reducing head training from hours to about twenty seconds and allowing the thirty training runs behind the learning curve. Second, the flattened pixel matrix is 3.2 GB in float32 and scikit-learn's brute-force k-NN drove the machine into swap, while libsvm requires float64 (6.4 GB) — almost certainly why the original authors capped their SVM at 3,000 images. Since k-NN and the RBF SVM need only pairwise distances, we compute these on the GPU in blocks and pass scikit-learn a precomputed kernel, which let us run the SVM on the full training set. The GPU path is checked against scikit-learn before use and agrees exactly for k-NN and to within 1e-5 relative error for the kernel; obtaining that precision required disabling TF32 and mean-centring the data, since the identity $||a-b||^2 = ||a||^2 + ||b||^2 - 2a\cdot b$ loses significance when the terms are five times larger than the result.

Every result is produced by a seeded script and written to JSON; this document is generated from those files, so its numbers cannot drift from the experiments. Verification scripts for label polarity and numerical agreement are part of the repository.

6. Conclusions

The paper's qualitative conclusion is confirmed: convolutional features transfer usefully to structural damage recognition, and classical models on raw pixels do not work at all on this task. Our quantitative results track the paper's own measured numbers closely, including a confusion-matrix accuracy within 0.1 points, although they fall short of its headline 83.0% — a figure inconsistent with its own error analysis. We disagree with the paper's bias diagnosis and present a seeded learning curve with error bars in support. The principal practical lessons were that the published data required two undocumented corrections before any model could be trained correctly, that neither error was detectable from accuracy alone, and that worthwhile accuracy remained available from simply matching the pretrained network's input resolution. Remaining work is MobileNetV1, the SVM on mid-network activations, and the fine-tuning and augmentation experiments reported on the authors' poster.

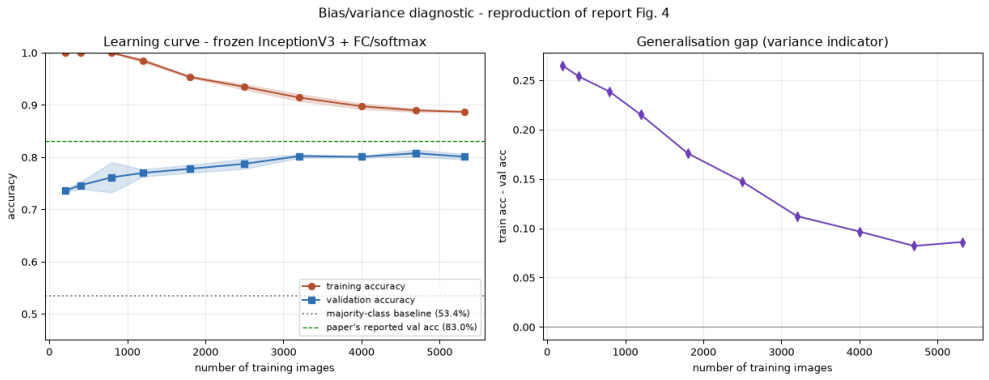


Figure 1. Learning curve for the frozen InceptionV3 model, mean of three seeds with shaded spread. Left: training and validation accuracy against training-set size. Right: the generalisation gap. The gap falls from 26 points to 8.6 as data is added while validation accuracy flattens near 80%, indicating a feature and label-noise limit rather than the bias the paper diagnosed.

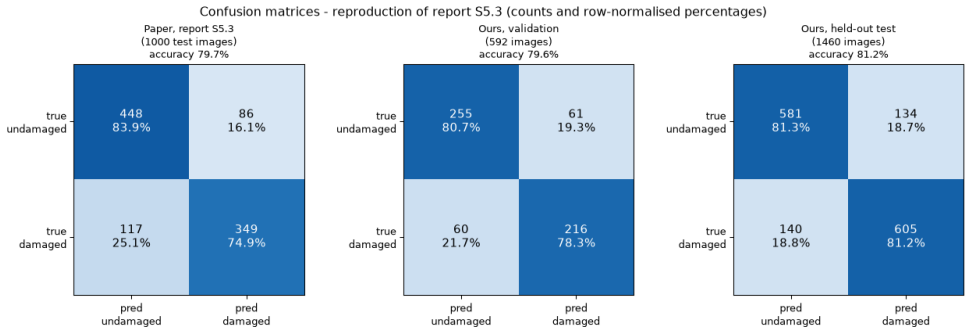


Figure 2. Confusion matrices, counts and row-normalised percentages. Left: the paper's, from its error-analysis section. Centre and right: ours on validation and on the held-out test set the paper did not keep. Our damaged-class recall (78.3% and 81.2%) exceeds the paper's 74.9%, which matters most for a triage application.